# Vectors I Homework 2: Scalar Product — Solutions

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True/False answers from the Basics Worksheet. Past-paper solutions: official school mark schemes where available; otherwise worked solutions (equation working). No cross-product methods used in any solution.

**1** For the following statements, state whether it is **true** or **false**.

| | **Statement** | **T/F** |
|---|---|---|
| (a) | $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$.<br>$a_1b_1+a_2b_2+a_3b_3=b_1a_1+b_2a_2+b_3a_3$ | **T** |
| (b) | $\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$.<br>$\mathbf{a}\cdot\mathbf{a}=a_1^2+a_2^2+a_3^2=\lvert\mathbf{a}\rvert^2$ | **T** |
| (c) | $\mathbf{i}\cdot\mathbf{j}=0$.<br>$1\cdot0+0\cdot1+0\cdot0=0$ | **T** |
| (d) | $\mathbf{a}\cdot\mathbf{b}$ is a vector.<br>$\begin{pmatrix}1\\0\\0\end{pmatrix}\cdot\begin{pmatrix}1\\0\\0\end{pmatrix}=1$ (a scalar) | **F** |
| (f) | If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$.<br>$\mathbf{0}\cdot\mathbf{b}=0b_1+0b_2+0b_3=0$ | **T** |
| (g) | $(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$.<br>$(2\mathbf{a})\cdot(3\mathbf{b})=2\cdot3\,(\mathbf{a}\cdot\mathbf{b})$ | **T** |
| (h) | $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector.<br>$(\mathbf{a}\cdot\mathbf{b})\mathbf{c}=\lambda\mathbf{c},\ \lambda=\mathbf{a}\cdot\mathbf{b}\in\mathbb{R}$ | **T** |
| (i) | $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$, where $\theta$ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$.<br>Definition of scalar product | **T** |
| (j) | $\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}$.<br>$\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=2+6+15=23$ (a scalar) | **F** |
| (k) | If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.<br>$\mathbf{i}\cdot\mathbf{j}=0,\ \mathbf{i}\neq\mathbf{0},\ \mathbf{j}\neq\mathbf{0}$ | **F** |
| (l) | $\lvert\mathbf{a}\cdot\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.<br>$\lvert\mathbf{i}\cdot\mathbf{j}\rvert=0\neq1=\lvert\mathbf{i}\rvert\lvert\mathbf{j}\rvert$ | **F** |
| (m) | $\mathbf{a}\cdot\mathbf{b}\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.<br>$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta\leq\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$ | **T** |

| | **Statement** | **T/F** |
|---|---|---|
| (n) | For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute.<br>$\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta>0 \therefore \cos\theta>0,\ 0^\circ\le\theta<90^\circ$ | **T** |
| (o) | For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is obtuse.<br>$\cos\theta<0 \therefore 90^\circ<\theta\le180^\circ$ | **T** |
| (q) | $\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar.<br>$\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore \mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is meaningless | **F** |
| (r) | $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+\lvert\mathbf{b}\rvert^2$.<br>$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+2\mathbf{a}\cdot\mathbf{b}+\lvert\mathbf{b}\rvert^2;\ \mathbf{a}=\mathbf{b}=\mathbf{i}: 4\neq2$ | **F** |
| (s) | For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel.<br>$\cos\theta=1 \therefore \theta=0^\circ$ | **T** |
| (t) | $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$ also holds if $\theta$ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$.<br>$\cos(360^\circ-\theta)=\cos\theta$ | **T** |
| (u) | Let $\theta$ be the angle $ABC$. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=\lvert\overrightarrow{BA}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$.<br>$\overrightarrow{BA}$, $\overrightarrow{BC}$ both start at $B$ | **T** |
| (v) | The length of projection of $\mathbf{a}$ onto $\mathbf{b}$ is equal to the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.<br>$\mathbf{a}=\mathbf{i},\ \mathbf{b}=2\mathbf{i}: \lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert=1\neq2=\lvert\mathbf{b}\cdot\hat{\mathbf{a}}\rvert$ | **F** |
| (w) | If $\lvert\mathbf{a}\rvert=\lvert\mathbf{b}\rvert$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$.<br>$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=\lvert\mathbf{a}\rvert^2-\lvert\mathbf{b}\rvert^2=0$ | **T** |
| (x) | $\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$.<br>$\mathbf{a}=\mathbf{i},\ \mathbf{b}=\mathbf{j},\ \mathbf{c}=\mathbf{k}: \mathbf{a}\cdot\mathbf{b}=0=\mathbf{a}\cdot\mathbf{c},\ \mathbf{b}\neq\mathbf{c}$ | **F** |
| (y) | Let $\theta$ be the angle $ABC$. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$.<br>$\overrightarrow{AB}\cdot\overrightarrow{BC}=-\overrightarrow{BA}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$ | **F** |
| (z) | Let $\theta$ be the angle $CBD$. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$.<br>$\overrightarrow{BD}\cdot\overrightarrow{CB}=-\overrightarrow{BD}\cdot\overrightarrow{BC}=-\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$ | **F** |

**2** [EJC 2024 Promo Q5]

*Worked solution (no official student script located; working matches Vectors I Test Review).*

**(a)**

$$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\frac{5\pi}{6}=\sqrt{2}\cdot\sqrt{6}\cdot\left(-\frac{\sqrt{3}}{2}\right)=-3$$

$$\text{Length of projection of } \mathbf{a} \text{ onto } \mathbf{b}=\frac{\lvert\mathbf{a}\cdot\mathbf{b}\rvert}{\lvert\mathbf{b}\rvert}=\frac{\lvert-3\rvert}{\sqrt{6}}=\frac{3}{\sqrt{6}}=\frac{\sqrt{6}}{2}.$$

**(b)**

$$\begin{aligned}(3\mathbf{a}+2\mathbf{b})\cdot(3\mathbf{a}+2\mathbf{b})&=9\lvert\mathbf{a}\rvert^2+12\mathbf{a}\cdot\mathbf{b}+4\lvert\mathbf{b}\rvert^2\\&=9(2)+12(-3)+4(6)=18-36+24=6\end{aligned}$$

$$\lvert3\mathbf{a}+2\mathbf{b}\rvert=\sqrt{6}.$$

**3** [SAJC 2024 Promo Q9(a)]

*Worked solution (no official student script located).*

**(i)**

$$\begin{aligned}(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b}) &= |2\mathbf{a}+\mathbf{b}|^2\\ 4|\mathbf{a}|^2+4\mathbf{a}\cdot\mathbf{b}+|\mathbf{b}|^2 &= (2\sqrt{74})^2 = 296\\ 4(81)+4\mathbf{a}\cdot\mathbf{b}+1 &= 296\\ 325+4\mathbf{a}\cdot\mathbf{b} &= 296\\ 4\mathbf{a}\cdot\mathbf{b} &= -29\\ \mathbf{a}\cdot\mathbf{b} &= -\frac{29}{4} \quad \text{(shown).}\end{aligned}$$

**(ii)**

Since $|\mathbf{b}|=1$, $|\mathbf{a}\cdot\mathbf{b}|$ is the length of the projection of $\mathbf{a}$ onto $\mathbf{b}$.

**(iii)**

$$\begin{aligned}\overrightarrow{PQ} &= (6\mathbf{a}+5\mathbf{b})-(7\mathbf{a}-5\mathbf{b}) = -\mathbf{a}+10\mathbf{b}\\ \overrightarrow{PR} &= (9\mathbf{a}+\lambda\mathbf{b})-(7\mathbf{a}-5\mathbf{b}) = 2\mathbf{a}+(\lambda+5)\mathbf{b}.\end{aligned}$$

Since $P$, $Q$, $R$ are collinear and $\mathbf{a}$, $\mathbf{b}$ are non-parallel, $\overrightarrow{PR}=k\overrightarrow{PQ}$ for some scalar $k$:

$$\begin{aligned}2\mathbf{a}+(\lambda+5)\mathbf{b} &= k(-\mathbf{a}+10\mathbf{b})\\ 2=-k, &\qquad \lambda+5=10k\\ k=-2, &\qquad \lambda+5=-20\\ \lambda=-25.&\end{aligned}$$

**4** [CJC 2025 Promo Q6(a)]

*Official solution.*

**(a)**

$$\begin{aligned}\overrightarrow{AC}=\mathbf{c}-\mathbf{a}, &\qquad \overrightarrow{BC}=\mathbf{c}-\mathbf{b}\\ \overrightarrow{AC}\cdot\overrightarrow{BC} &= (\mathbf{c}-\mathbf{a})\cdot(\mathbf{c}-\mathbf{b}) = (\mathbf{c}-\mathbf{a})\cdot(\mathbf{c}-(-\mathbf{a})) = (\mathbf{c}-\mathbf{a})\cdot(\mathbf{c}+\mathbf{a})\\ &= |\mathbf{c}|^2-|\mathbf{a}|^2 = 0 \quad \text{(since } |\mathbf{a}|=|\mathbf{c}|=\text{radius).}\end{aligned}$$

Hence $AC$ and $BC$ are perpendicular (shown).

**5** [CJC 2022 Promo Q9(i),(ii)]

*Official solution (CJC markers' report).*

**(i)**

$$\begin{aligned}\mathbf{b}\cdot(\mathbf{b}-\mathbf{a}) &= 0\\ |\mathbf{b}|^2-\mathbf{a}\cdot\mathbf{b} &= 0\\ \mathbf{a}\cdot\mathbf{b} &= 1 \qquad \text{(since } \mathbf{b} \text{ is a unit vector)}\\ |\mathbf{a}||\mathbf{b}|\cos\theta &= 1\\ |\mathbf{a}|\cos\theta &= 1\\ |\mathbf{a}| &= \sec\theta \quad \text{(shown).}\end{aligned}$$

**(ii)**

By the ratio theorem, $\mathbf{c}=\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}$.

$$\mathbf{c}\cdot\mathbf{b}=\left(\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}\right)\cdot\mathbf{b}=\frac{1}{4}\mathbf{a}\cdot\mathbf{b}+\frac{3}{4}|\mathbf{b}|^2=\frac{1}{4}(1)+\frac{3}{4}=1$$
$$|\mathbf{c}\cdot\mathbf{b}|=1.$$

$|\mathbf{c}\cdot\mathbf{b}|$ is the length of the projection of $\mathbf{c}$ onto $\mathbf{b}$.

**6** [SAJC 2025 Promo Q8(i)]

*Worked solution (no official student script located).*

**(i)**

$$(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b})=|3\mathbf{a}-2\mathbf{b}|^2$$
$$9|\mathbf{a}|^2-12\mathbf{a}\cdot\mathbf{b}+4|\mathbf{b}|^2=37$$
$$9(4)-12\mathbf{a}\cdot\mathbf{b}+4(1)=37$$
$$40-12\mathbf{a}\cdot\mathbf{b}=37$$
$$\mathbf{a}\cdot\mathbf{b}=\frac{1}{4}\quad\text{(shown).}$$

$|\mathbf{a}\cdot\hat{\mathbf{b}}|$ is the length of the projection of $\mathbf{a}$ onto $\mathbf{b}$.

**7** [EJC 2025 Prelim P1 Q6]

*Official solution.*

**(a)**

$$\overrightarrow{AP}=\mathbf{p}-\mathbf{a},\qquad \overrightarrow{BP}=\mathbf{p}+\mathbf{a}\quad(\text{since }\overrightarrow{OB}=-\mathbf{a})$$
$$\overrightarrow{AP}\cdot\overrightarrow{BP}=(\mathbf{p}-\mathbf{a})\cdot(\mathbf{p}+\mathbf{a})=|\mathbf{p}|^2-|\mathbf{a}|^2=0\quad(\text{since }|\mathbf{a}|=|\mathbf{p}|=\text{radius}).$$

**(b)**

Since $AC:AB=\lambda:1$ and $\overrightarrow{AB}=-2\mathbf{a}$,

$$\overrightarrow{AC}=\lambda\,\overrightarrow{AB}=-2\lambda\mathbf{a}$$
$$\mathbf{c}=\mathbf{a}+\overrightarrow{AC}=(1-2\lambda)\mathbf{a}.$$

**(c)**

Since $PC$ is a tangent, $\overrightarrow{OP}\perp\overrightarrow{PC}$:

$$\mathbf{p}\cdot((1-2\lambda)\mathbf{a}-\mathbf{p})=0$$
$$(1-2\lambda)\,\mathbf{p}\cdot\mathbf{a}-|\mathbf{p}|^2=0$$
$$(1-2\lambda)\,|\mathbf{a}||\mathbf{p}|\cos 120^\circ-|\mathbf{a}|^2=0$$
$$(1-2\lambda)\,|\mathbf{a}|^2\left(-\frac{1}{2}\right)-|\mathbf{a}|^2=0$$
$$-\frac{1}{2}(1-2\lambda)=1$$
$$1-2\lambda=-2$$
$$\lambda=\frac{3}{2}.$$

**8** [NJC 2025 Prelim P2 Q3]

*Official solution (Method 2: scalar multiple; no cross product used).*

**(i)**

Since $O$, $P$, $Q$ are collinear, $\mathbf{q} = k\mathbf{p}$ for some scalar $k$:

$$\begin{pmatrix} \cos 2t \\ -\sin 2t \\ \alpha \end{pmatrix} = k \begin{pmatrix} \cos t \\ \sin t \\ -1 \end{pmatrix}.$$

From the **j**-component:

$$\begin{aligned} -\sin 2t &= k \sin t \\ -2\cos t \sin t &= k \sin t \\ \sin t(-2\cos t - k) &= 0 \end{aligned}$$

$$\sin t = 0 \quad \text{(reject, since } 0 < t < \pi\text{)} \quad \text{or} \quad \cos t = -\frac{k}{2}.$$

From the **i**-component:

$$\begin{aligned} \cos 2t &= k \cos t \\ 2\cos^2 t - 1 - k\cos t &= 0. \end{aligned}$$

Substitute $\cos t = -\frac{k}{2}$:

$$\begin{aligned} 2\left(-\frac{k}{2}\right)^2 - k\left(-\frac{k}{2}\right) - 1 &= 0 \\ \frac{k^2}{2} + \frac{k^2}{2} - 1 &= 0 \\ k^2 &= 1 \\ k &= \pm 1. \end{aligned}$$

From the **k**-component, $\alpha = -k$.

$$k = 1 \Rightarrow \cos t = -\frac{1}{2} \Rightarrow t = \frac{2\pi}{3} \ (0 < t < \pi), \quad \alpha = -1$$

$$k = -1 \Rightarrow \cos t = \frac{1}{2} \Rightarrow t = \frac{\pi}{3}, \quad \alpha = 1.$$

Hence $t = \frac{\pi}{3}$, $\alpha = 1$ or $t = \frac{2\pi}{3}$, $\alpha = -1$.

**(ii)**

$$\mathbf{p} \cdot \mathbf{q} = \cos t \cos 2t - \sin t \sin 2t - 0.5 = \cos(2t + t) - 0.5 = \cos 3t - 0.5 \quad \text{(shown)}.$$

$$|\mathbf{p}| = \sqrt{\cos^2 t + \sin^2 t + 1} = \sqrt{2}$$

$$\text{Length of projection of } \mathbf{q} \text{ onto } \mathbf{p} = \frac{|\mathbf{p} \cdot \mathbf{q}|}{|\mathbf{p}|} = \frac{|\cos 3t - 0.5|}{\sqrt{2}}.$$

The greatest value of $|\cos 3t - 0.5|$ on $0 < t < \pi$ is 1.5 (when $\cos 3t = -1$), so

$$\text{greatest length} = \frac{1.5}{\sqrt{2}} = \frac{3}{2\sqrt{2}} \quad (= 1.06 \text{ (3 s.f.)}).$$

**END OF SOLUTIONS**